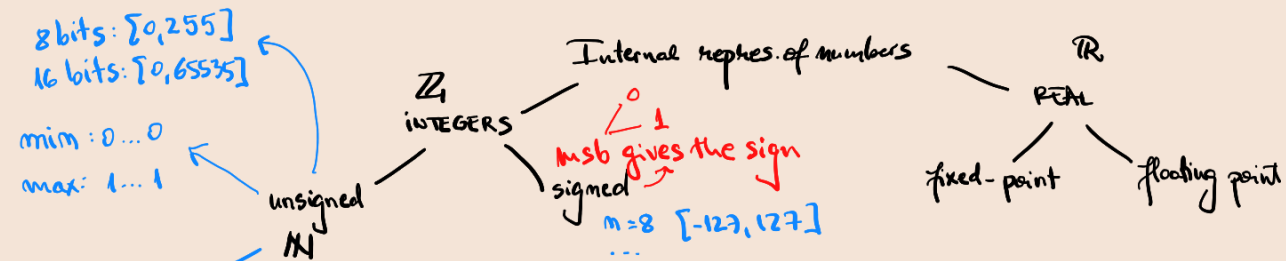
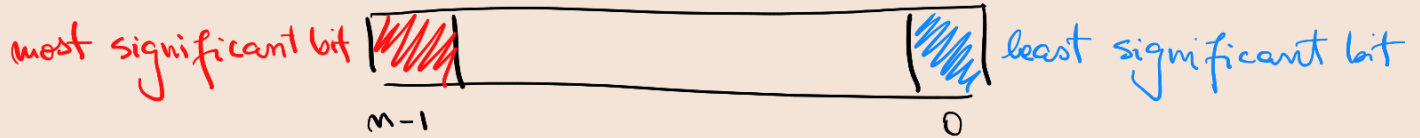


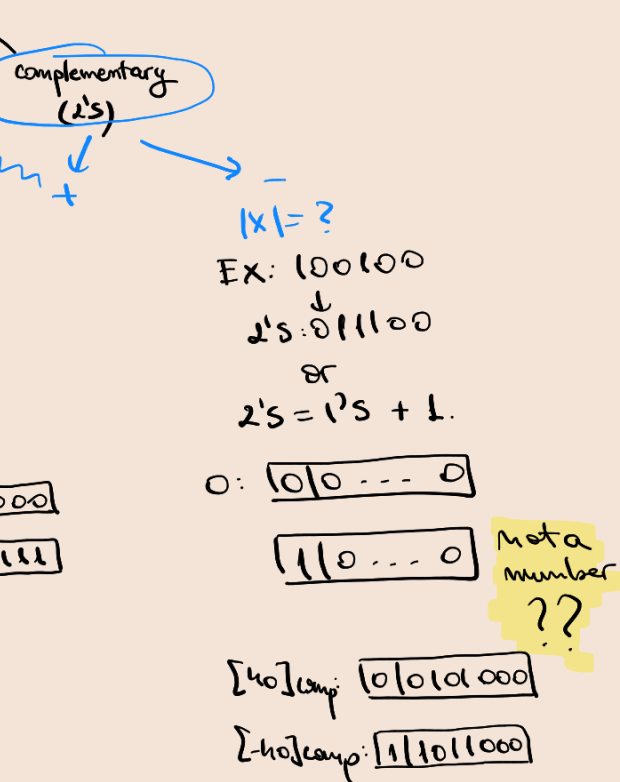
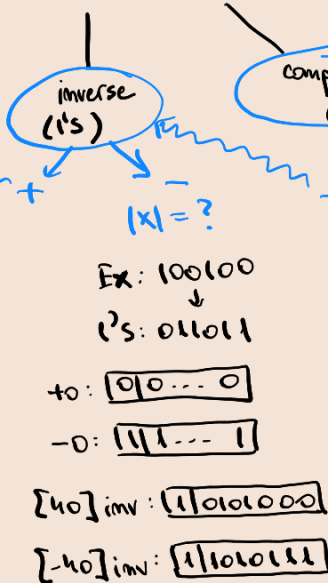
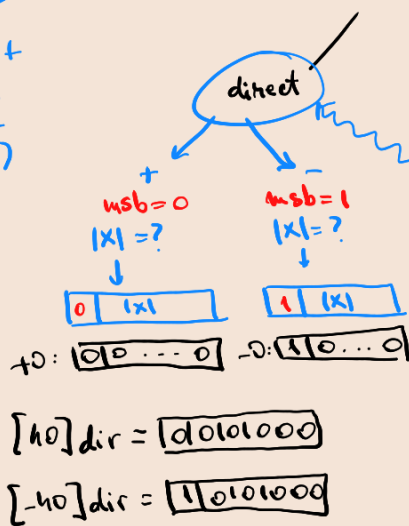
Lecture 2 (9.10.2025)

$m \in \{8, 16, 32, 64, \dots\}$ bits MEMORY DIMENSION



$1111110011 (2) +$
 $00010010 (2)$

 $1000010101 (2)$
 overflow



$$x, y \in \mathbb{Z}_+$$

$$[x+y]_{comp} = [x]_{comp} + [y]_{comp}$$

$$[x-y]_{comp} = [x]_{comp} + [-y]_{comp}$$

Algebraic sum = true signed sum.

$$sgn(x_1) = sgn(x_2) = s$$

$$sgn([x_1]_{comp} + [x_2]_{comp}) \neq s$$

$\Rightarrow$ overflow

- Add/sub of integers in complementary code:

$$[40+40]_{\text{comp}} = [40]_{\text{comp}} + [40]_{\text{comp}}$$

$$\begin{array}{r} \boxed{010101000} (40_2) \\ \boxed{010101000} (40_2) \\ \hline \boxed{011010000} \end{array}$$

$$[80]_{\text{comp}} \Rightarrow \boxed{011010000}$$

$$[100+100]_{\text{comp}} = [200]_{\text{comp}}$$

(r1) 100, 100 have the same sign, but 100+100 has the op. sign. $\Rightarrow$ overflow

$$\begin{array}{r} \boxed{01100100} (100_2) \\ \boxed{01100100} (100_2) \\ \hline \boxed{111001000} \text{ overflow} \end{array}$$

$$[100-40]_{\text{comp}} = [100]_{\text{comp}} + [-40]_{\text{comp}}$$

$$\begin{array}{r} \boxed{01100100} (100_2) \\ \boxed{11011000} (-40_2) \\ \hline \boxed{10011100} \end{array}$$

$$[60]_{\text{comp}} \Rightarrow \boxed{10011100} \checkmark$$

(r2) if there is a carry digit outside $\rightarrow$ this will be eliminated

If there is no overflow we interpret the content of the memory location as a complementary code of an int number. $\rightarrow$ 2's comp.

Subunitary numbers

$m=8$ bits

$$6/8 = \frac{6}{8} = 6 \cdot 8^{-1} = 0,6(8) = 0,110(2)$$

$$X = 11/16 = 11 \cdot 16^{-1} = 0,13(16) = 0,1011(2)$$

$$y = 0,45 = 0,346(8) = 0,01110011(2)$$

$$0,45 \cdot 8 = 3,6$$

$$0,6 \cdot 8 = 4,8$$

$$0,8 \cdot 8 = 6,4$$

$8=2^3 \Rightarrow$ groups of 3

$$[11/16]_{\text{dir}} = [11/16]_{\text{inv}} = [11/16]_{\text{comp}} = \boxed{0110110000}$$

$$[11/16]_{\text{dir}} = \boxed{1110110000}$$

$$[-11/16]_{\text{dir}} = \boxed{1 \mid 1 \mid 0 \mid 1 \mid 1 \mid 1 \mid 1 \mid 1}$$

$$[-11/16]_{\text{inv}} = \boxed{1 \mid 0 \mid 1 \mid 1 \mid 0 \mid 1 \mid 1 \mid 1}$$

$$[-11/16]_{\text{comp}} = \boxed{1 \mid 0 \mid 1 \mid 1 \mid 0 \mid 1 \mid 0 \mid 0}$$

$$[0,45]_{\text{dir}} = [0,45]_{\text{inv}} = [0,45]_{\text{comp}} = \boxed{0 \mid 0 \mid 1 \mid 1 \mid 1 \mid 0 \mid 1 \mid 0}$$

$$\begin{matrix} [-0,45]_{\text{dir}}, & [-0,45]_{\text{inv}}, & [-0,45]_{\text{comp}} \end{matrix}$$

$$[11/16 + 0,45]_{\text{comp}} = [11/16]_{\text{comp}} \oplus [0,45]_{\text{comp}}$$

⊕

$$[11/16]_{\text{comp}} : \boxed{0 \mid 1 \mid 0 \mid 1 \mid 1 \mid 0 \mid 0 \mid 0}$$

$$[0,45]_{\text{comp}} : \boxed{0 \mid 0 \mid 1 \mid 1 \mid 1 \mid 0 \mid 0 \mid 1}$$

$$\boxed{1 \mid 1 \mid 0 \mid 1 \mid 0 \mid 0 \mid 0 \mid 1} \quad \oplus$$

↓
overflow

$$\ominus \quad [11/16]_{\text{comp}} : \boxed{0 \mid 1 \mid 0 \mid 1 \mid 1 \mid 0 \mid 0 \mid 0}$$

$$[-0,45]_{\text{comp}} : \boxed{1 \mid 1 \mid 0 \mid 0 \mid 0 \mid 1 \mid 1 \mid 1}$$

$$\boxed{1 \mid 0 \mid 0 \mid 0 \mid 1 \mid 1 \mid 1 \mid 1} \quad \oplus \quad \text{correct result} \checkmark$$

$$[0,45 - 11/16]_{\text{comp}} = [0,45]_{\text{comp}} + [-11/16]_{\text{comp}}$$

$$[-11/16]_{\text{comp}} \quad \boxed{1 \mid 0 \mid 1 \mid 0 \mid 1 \mid 1 \mid 0 \mid 0}$$

$$[0,45]_{\text{comp}} \quad \boxed{0 \mid 0 \mid 1 \mid 1 \mid 1 \mid 0 \mid 0 \mid 1}$$

$$\boxed{1 \mid 1 \mid 1 \mid 0 \mid 0 \mid 0 \mid 0 \mid 1} \quad +$$

2's comp:

0 0 0 1 1 1 1 1

$$13/16 = 0, D(16) = 0, 1101(2)$$

$$13/16 > 0 \Rightarrow [13/16]_{dir} = [13/16]_{inv} = [13/16]_{comp} =$$

0 1 1 0 1 0 0 0

$[-13/16]_{dir} = 1 1 1 1 0 1 0 0$

$[-13/16]_{inv} = 1 1 0 0 1 1 0 1 1 1$

$[-13/16]_{comp} = 1 0 0 1 1 0 0 0$

$$0,25 = 0,01(2)$$

$[0,25]_{comp} = 1 0 0 1 0 0 0 0$

$[-0,25]_{comp} = 1 1 1 1 0 0 0 0$

I = ... bits; F = ... bits

Fixed-point representation

$$-29,21 \rightarrow 16 \text{ bits, } I = 6 \text{ bits, } F = 9 \text{ bits}$$

$$29 = 11101(2)$$

$$0,21 \cdot 2 = 0,42$$

$$0,42 \cdot 2 = 0,84$$

$$0,84 \cdot 2 = 1,68$$

$$0,68 \cdot 2 = 1,36$$

$$0,36 \cdot 2 = 0,72$$

1

0

1

1

1 0 1 1 1 0 1 0 0 1 1 0 1 0 1 1 (*)

↓
" "

I = 6

F = 9

$$2 \rightarrow 2^4 = 16 \Rightarrow \text{groups of 4.}$$

↓
B

↓
A

↓
G

↓
B

$$\Rightarrow -29,21(10) = B A G B(16) = * (2)$$

$$0,21 = 0,001101011(2)$$

$$-29,21 = -11101,001101011(2)$$

Floating point representation of real numbers s, exp, mantissa

Mantissa

$$X = \pm 0, m \cdot b^e$$

$\downarrow$ mantissa
 $\nearrow$ exponent
 $\searrow$ numeration base

$$X = \pm 1, m \cdot b^e$$